Golden-Ratio Non-Resonance in Phase Space as an Organizing Hypothesis for Dark-Sector Anomalies

Pre-registered Predictions for the Dark-Energy Equation of State, the Growth Index and Early Dark Energy

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Abstract

We propose that the departures of the dark sector from ΛCDM are set by one number, the inverse golden ratio $x = 1/\varphi = (\sqrt{5}-1)/2$. The proposal has two strictly separated parts. The mathematical part is proven, apart from one labeled numerical result: distinct phase-space states never merge; $1/\varphi$ is the global attractor of the self-similar partition $x = 1/(1+x)$ and, by Hurwitz's theorem, the most badly approximable number; and golden KAM tori are found numerically to be the most robust barriers in model four-dimensional phase spaces. The physical part is a hypothesis: dark-sector parameters take ladder values $\alpha_d x^d$, with $\alpha_d = 1$ or $\frac{1}{2}$. Without free parameters it gives $w_0 = -\varphi/2$, $w_a = -1/\varphi$, a crossing of $w = -1$ at $z = 1/\sqrt{5}$, a growth index $\gamma = 1/\varphi$ and an early-dark-energy fraction $f_{\rm EDE} = x^4/2 = 0.073$. With DESI DR2 BAO, CMB data and four supernova samples, $(w_0, w_a) = (-0.809, -0.618)$ lies 0.3–1.0σ from the free fit, has the lowest AIC and BIC of the models tested, and lies at 0.5–1.0σ in DESI's official posteriors with supernovae. Of 403 pairs from the same algebraic family, it alone is within 1σ for all three main supernova samples. The agreement holds without CMB data and with the predicted early dark energy. Fitting x freely gives $x = 0.55$–0.69: consistent with $1/\varphi$, but neighboring values fit almost as well. The growth index is 1.1–1.3σ from CMB-anchored $f\sigma_8$ data. An earlier claim of $H_0 = 72.7\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is withdrawn, and momentum exchange is shown not to produce the predicted growth suppression. All predictions are time-stamped before any DESI data beyond DR2.

1 Introduction

Three observational tensions currently challenge ΛCDM:

1. **Evolving dark energy.** DESI DR2 baryon acoustic oscillations (BAO), combined with the cosmic microwave background (CMB) and type Ia supernovae (SNe), prefer an evolving equation of state $w(a) = w_0 + w_a(1-a)$ with $w_0 > -1$ and $w_a < 0$, at 2.8–4.2σ depending on the SN sample [1]. A recalibrated DES sample (DES-Dovekie) gives a similar but weaker preference, 3.2σ, with $w_0 = -0.803 \pm 0.054$ and $w_a = -0.72 \pm 0.21$ from DESI DR2, CMB (Planck, ACT and SPT) and DES-Dovekie [4].

2. **Clustering amplitude.** Some weak-lensing surveys reported a lower S_8 than the CMB-anchored ΛCDM value [6], although KiDS-Legacy finds $S_8 = 0.815$, largely consistent with the CMB [7].

3. **The Hubble tension.** The distance ladder gives $H_0 = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [8], versus $67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from the CMB [5].

We ask whether a single geometric principle can fix the size of these departures *without free parameters*. Any such proposal faces the objection of numerology: with enough simple expressions, some will match any measurement. We address it in four ways:

1. We separate proven mathematics (Sec. 2) from the physical hypothesis (Sec. 3).
2. We quantify the look-elsewhere effect (Sec. 4.8).
3. We let the data measure x and compare it with other constants (Sec. 4.6).
4. We time-stamp all predictions and falsification criteria before new data (Sec. 5).

We also report, without omission, which earlier claims did not survive testing.

Related work. The golden ratio has appeared in exact solutions of general relativity, for example in the orbits of Schwarzschild–Kottler black holes [9] and in periodic Bianchi IX cosmologies [10]. A search of arXiv abstracts found no previous proposal that fixes the dark-energy equation of state or the growth index by the golden ratio.

2 Mathematical foundation

This section collects proven results (Propositions 1–3, Lemmas 1 and 2, and the theorems of Hurwitz and KAM). It also states one well-supported but not fully general result (Remark 2), which is labeled as such.

2.1 Distinct states never merge in phase space

Proposition 1 (Classical non-intersection)**.** *Let $\dot{\mathbf{X}} = \mathbf{V}(\mathbf{X})$ on phase space $\mathcal{P} \subset \mathbb{R}^n$ with $\mathbf{V}$ locally Lipschitz. If two solutions satisfy $\mathbf{X}_1(t_0) = \mathbf{X}_2(t_0)$ at any instant t_0, then $\mathbf{X}_1(t) = \mathbf{X}_2(t)$ on their common interval of existence. Distinct trajectories therefore never intersect.*

Proof. Both solve the same initial-value problem, whose solution is unique by the Picard–Lindelöf theorem [11]. □

For Hamiltonian flows, Liouville's theorem adds that phase-space volume is conserved.

Proposition 2 (Quantum non-merging)**.** *Let U be unitary. For any states $|\psi\rangle, |\chi\rangle$, $\langle U\psi|U\chi\rangle = \langle\psi|\chi\rangle$. Consequently:*

(i) orthogonal states remain orthogonal;

(ii) states with $|\langle\psi|\chi\rangle| < 1$ never evolve into the same state.

Proof. $\langle U\psi|U\chi\rangle = \langle\psi|U^\dagger U|\chi\rangle = \langle\psi|\chi\rangle$, because $U^\dagger U = 1$. □

Remark 1. Proposition 2 is the precise content of the intuition that "a collision in phase space would destroy information". It concerns distinct states; identical bosons may share a state [12], so no universal "no-collision" law for particles is implied. Only the classical results are used below.

2.2 Self-similar partition and the golden ratio

Suppose a region of phase space is partitioned into a mobile "escape" fraction x and an excluded "barrier" fraction $1 - x$, and that the partition is *self-similar*: the barrier-to-channel ratio equals the channel-to-whole ratio.

Lemma 1. *The condition* $(1-x)/x = x/1$ *with* $x > 0$ *has the unique solution*

$$x = \frac{\sqrt{5}-1}{2} = \frac{1}{\varphi} \approx 0.618034, \qquad \varphi = \frac{1+\sqrt{5}}{2}. \tag{1}$$

The condition is equivalent to $x = 1/(1+x)$*, so* $x = [0; 1, 1, 1, \dots]$ *is the continued fraction whose partial quotients are all equal to the minimal value* 1*.*

Proof. $(1-x)/x = x \iff x^2 + x - 1 = 0 \iff x(1+x) = 1 \iff x = 1/(1+x)$. The positive root is $(\sqrt{5}-1)/2$; the other root is $-\varphi$. Iterating $x = 1/(1+x)$ generates the continued fraction. □

Proposition 3 (The recursive partition converges to $1/\varphi$)**.** *Let* $f(x) = 1/(1+x)$ *and* $x_{n+1} = f(x_n)$ *with any* $x_0 \geq 0$*. Then*

$$|x_n - \varphi^{-1}| \leq \varphi^{-n}\, |x_0 - \varphi^{-1}|, \tag{2}$$

so $x_n \to 1/\varphi$ *geometrically. For* $x_0 = 1$*,* $x_n = F_{n+1}/F_{n+2}$ *(ratios of Fibonacci numbers).*

Proof. Write $x_* = 1/\varphi$, which satisfies $x_* = 1/(1+x_*)$. For $x \geq 0$,

$$|f(x) - x_*| = \frac{|x - x_*|}{(1+x)(1+x_*)} \leq \frac{|x-x_*|}{1+x_*} = x_*\, |x - x_*|. \tag{3}$$

Induction gives the bound. For the Fibonacci statement, $F_{n+1}/F_{n+2} = F_{n+1}/(F_{n+1}+F_n) = 1/(1+F_n/F_{n+1})$. □

So the golden ratio is not merely *a* self-similar solution: it is the global attractor of the recursive partition, reached from any starting point. The map f itself is not a contraction on $[0,\infty)$, since $|f'(0)| = 1$; the bound above contracts distances *to the fixed point*. The two-step map $f \circ f(x) = (1+x)/(2+x)$ is a contraction on $[0,\infty)$ with constant $1/4$, so the Banach fixed-point theorem applies to it.

2.3 Non-resonance: why $1/\varphi$ is special

Theorem 1 (Hurwitz 1891 [13])**.** *For every irrational* α *there are infinitely many rationals* p/q *with* $|\alpha - p/q| < 1/(\sqrt{5}\, q^2)$*. The constant* $\sqrt{5}$ *cannot be improved: for* $\alpha = 1/\varphi$ *and any* $c > \sqrt{5}$*, only finitely many* p/q *satisfy* $|\alpha - p/q| < 1/(c\, q^2)$*.*

Numbers equivalent to $1/\varphi$, that is, related to it by $\alpha \mapsto (a\alpha + b)/(c\alpha + d)$ with integers $ad - bc = \pm 1$, are exactly those whose continued fraction ends in an unbroken tail of 1s (the *noble* numbers). They are the *most badly approximable* by rationals: they stay as far as possible from every resonance p/q.

Theorem 2 (KAM, informal [14, 15, 16])**.** *In a nearly integrable Hamiltonian system with two degrees of freedom (a four-dimensional phase space) that satisfies the standard non-degeneracy condition (a twist condition for the associated area-preserving map), invariant tori whose frequency ratio is sufficiently badly approximable (Diophantine) survive sufficiently small smooth perturbations. On a three-dimensional energy surface these two-dimensional tori are impassable barriers to trajectories.*

KAM theory does *not* say that only golden tori survive; most sufficiently irrational tori do. Greene's residue criterion, a numerical method, and the renormalization-group analysis of MacKay indicate that in the standard map the invariant circle with golden-mean winding number is the *last* to break, at $K_c \approx 0.971635$ [17, 18]. Its break-up is self-similar under a renormalization operator. In phase spaces of six or more dimensions, tori no longer separate the energy surface, and trajectories can drift around them (Arnold diffusion [19]). The barrier picture is therefore specific to two degrees of freedom.

Remark 2 (Evidence, not a general theorem). Combining the results above: the winding ratio that best avoids resonance is $1/\varphi$ (Hurwitz, proven). In the standard map, and conjecturally in generic area-preserving maps, the corresponding golden invariant circle is the most robust barrier (Greene, MacKay: numerical and renormalization evidence). Its self-similar structure is the fixed point of Lemma 1 and the attractor of Proposition 3.

2.4 Algebraic identities used below

Lemma 2. *With $x = 1/\varphi$:*

(a) $x^2 = 1 - x$, $x^3 = 2x - 1 = \sqrt{5} - 2$ *and* $x^4 = 2 - 3x$;

(b) $1 - \frac{1}{2}x^2 = \frac{1}{2}(1 + x) = \varphi/2$;

(c) $\frac{1}{2}x^4 = (5 - 3\varphi)/2 \approx 0.072949$;

(d) $x/(2 - x) = 1/\sqrt{5} = 1/(2\varphi - 1)$;

(e) for any x, $-x - \big(1 + 2(-1 + \frac{1}{2}x^2)\big) = -(x^2 + x - 1)$.

Proof. (a) From $x^2 + x - 1 = 0$; then $x^3 = x - x^2 = 2x - 1$ and $x^4 = (1 - x)^2 = 2 - 3x$. (b) $1 - \frac{1}{2}(1 - x) = \frac{1}{2}(1 + x)$, and $1 + x = \varphi$. (c) $\frac{1}{2}(2 - 3x) = 1 - \frac{3}{2}(\varphi - 1) = (5 - 3\varphi)/2$. (d) $2 - x = \sqrt{5}(\sqrt{5} - 1)/2$; and $2\varphi - 1 = \sqrt{5}$. (e) Direct expansion. □

All identities in this paper were also verified as exact symbolic identities with SymPy (see Data and code availability).

3 The physical hypothesis

Everything in this section is *postulated*. Its justification is empirical (Sec. 4), and its weak points are listed in Sec. 6.

Hypothesis 1 (Golden non-resonance of the dark sector)**.** The large-scale dynamics of the dark sector is organized like the maximally non-resonant system of Sec. 2: departures from ΛCDM are fixed by the non-resonant fraction $x = 1/\varphi$.

Hypothesis 2 (Dimensional ladder)**.** A dark-sector quantity that involves d independent phase-space directions is set by the ladder value

$$\mathcal{O}_d = \alpha_d\, x^d, \qquad \alpha_d = \begin{cases} 1 & \text{linear or volumetric modes } (d = 1, 3), \\ \frac{1}{2} & \text{transverse or plasma (kinetic) modes } (d = 2, 4). \end{cases} \tag{4}$$

The factor x^d follows if the golden partition acts independently in each of d directions, as a product measure. The weights α_d are part of the hypothesis. The value $\frac{1}{2}$ is motivated by equipartition (one half per quadratic degree of freedom) but is not derived. The ladder values are $\mathcal{O}_1 = 0.618034$, $\mathcal{O}_2 = 0.190983$, $\mathcal{O}_3 = 0.236068$ and $\mathcal{O}_4 = 0.072949$.

Hypothesis 3 (Assignment)**.**

1. Late-time dark energy is a transverse $d = 2$ departure: $1 + w_0 = \mathcal{O}_2$.
2. Structure growth is a linear $d = 1$ mode: $\gamma = \mathcal{O}_1$.
3. The pre-recombination early-dark-energy fraction is a plasma $d = 4$ departure: $f_{\rm EDE} = \mathcal{O}_4$.
4. The evolution parameter is $w_a = -x$. Equivalently, the crossing of $w = -1$ lies at $1 - a_{\rm cross} = x/2$, so that $w_a = -(1 + w_0)/(1 - a_{\rm cross}) = -x$. This is an equivalent postulate, not a derivation (Open problem 3).

For dark energy and early dark energy the ladder value is the departure from ΛCDM itself: $1 + w_0$, w_a and $f_{\rm EDE}$ all vanish in ΛCDM. For growth it is the value of γ, not its shift from the general-relativity value $\gamma_{\rm GR} \simeq 0.55$; a shift-type rule is discussed in Remark 3.

3.1 Consequences for dark energy

Using the Chevallier–Polarski–Linder (CPL) form $w(a) = w_0 + w_a(1-a)$ [20, 21], a local (first-order in $1-a$) parametrization that we test below also without assuming it (Appendix A), and Lemma 2:

$$w_0 = -1 + \tfrac{1}{2}x^2 = -\frac{\varphi}{2} = -0.80902, \tag{5}$$

$$w_a = -x = -\frac{1}{\varphi} = -0.61803, \tag{6}$$

$$w_a = 1 + 2w_0, \tag{7}$$

$$1 + w_0 + w_a = \tfrac{1}{2}(1-3x) = -0.42705. \tag{8}$$

By Lemma 2(e), the linear relation (7) holds on the curve $(w_0, w_a) = (-1+x^2/2, -x)$ *only* at the golden value, so it is an equivalent, data-friendly statement of $x^2 + x - 1 = 0$. The dark-energy density evolves as

$$\frac{\rho_{\rm DE}(a)}{\rho_{\rm DE,0}} = a^{-3(1+w_0+w_a)}\, e^{-3w_a(1-a)} = a^{\frac{3}{2}(3x-1)}\, e^{3x(1-a)}. \tag{9}$$

The equation of state crosses $w = -1$ where $z/(1+z) = (1+w_0)/(-w_a) = x/2$, that is,

$$z_{\rm cross} = \frac{x}{2-x} = \frac{1}{\sqrt{5}} = \frac{1}{2\varphi - 1} = 0.44721. \tag{10}$$

At fixed redshifts the prediction is

$$w(z=0.3) = -\frac{7+19\sqrt{5}}{52} = -0.9516, \qquad w(z=0.5) = -\frac{1+5\sqrt{5}}{12} = -1.0150. \tag{11}$$

Physical meaning of the crossing. Because $\mathrm{d}\ln\rho_{\rm DE}/\mathrm{d}\ln a = -3(1+w)$, the dark-energy density is *maximal* exactly at $z_{\rm cross}$, about 4.75 Gyr ago in the best-fit background. Since then $w > -1$, so dark energy is diluting and there is no "Big Rip". A history with $w_0 > -1$ and $w_0 + w_a < -1$, phantom in the past and quintessence-like today, is called *quintom-B* [22]. The present values satisfy both conditions ($w_0 + w_a = -1.427$), and this is the region favored by DESI [1, 22]. In the same background, dark energy overtakes matter at $z = 0.334$ and acceleration begins at $z = 0.729$; neither coincides with $z_{\rm cross}$.

Future evolution (qualitative only). Extrapolating the CPL form, acceleration would end when the universe is 1.67 times its present size, and matter would dominate again at 5.2 times. However, the same extrapolation gives $w > 1$ beyond 3.9 times the present size, which is unphysical, so only the qualitative statement survives: in this model acceleration is a transient phase, unlike for a cosmological constant.

3.2 Growth of structure

Write the linear growth rate as $f(z) = \Omega_{\rm m}(z)^{\gamma}$. In general relativity (GR), $\gamma_{\rm GR} \simeq 3(1-w)/(5-6w)$ for slowly varying w, giving $6/11$ for $w = -1$ [23, 24]. Hypothesis 3 predicts a constant

$$\gamma = x = 1/\varphi = 0.618. \tag{12}$$

Remark 3 (An alternative growth rule). An earlier variant set $\gamma = \gamma_{\rm GR}(z) + \mathcal{O}_4$, a $d = 4$ shift on top of GR. In the best-fit background this gives $\gamma = 0.629, 0.620, 0.617$ at $z = 0, 0.5, 1$, numerically almost identical to $1/\varphi$. The near-equality follows from the identity $\frac{6}{11} + \mathcal{O}_4 - \frac{1}{\varphi} = \frac{5}{2}\left(\frac{89}{55} - \varphi\right) = 3.7 \times 10^{-4}$, in which $89/55$ is a Fibonacci convergent of φ. That variant implies a consistency relation $\gamma - \gamma_{\rm GR} = f_{\rm EDE}$. We adopt $\gamma = x$ because it follows the ladder assignment (growth is a linear, $d = 1$ mode); current data cannot distinguish the two forms ($\chi^2 = 8.13$ versus 8.19, Sec. 4.9).

3.3 Embedding in standard frameworks

The ladder fixes numbers, not dynamics. To make the predictions computable and comparable with published constraints, we embed each of them in an established effective framework.

Phantom crossing: parametrized post-Friedmann dark energy. A single canonical scalar field cannot cross $w = -1$ [25]. The standard remedy, used by DESI whenever w crosses -1 [1], is the parametrized post-Friedmann (PPF) description [26]. It treats dark energy as an effective component whose perturbations remain stable through the crossing. Prediction 1 is a statement about this effective $w(a)$.

Growth: effective gravitational coupling. In the quasi-static regime, phenomenological modified gravity multiplies the source of matter clustering by $\mu(a)$ [27]. With $N = \ln a$ and $f = \mathrm{d}\ln\delta/\mathrm{d}N$,

$$f' + f^2 + \left(2 + \frac{\mathrm{d}\ln H}{\mathrm{d}N}\right) f = \frac{3}{2}\,\mu(a)\,\Omega_{\mathrm{m}}(a). \tag{13}$$

Inserting $f = \Omega_{\mathrm{m}}^{1/\varphi}$ in the best-fit φ-ladder background gives $\mu - 1 = -0.178$, -0.125, -0.068 and -0.020 at $z = 0$, 0.5, 1 and 2. This is described to within 0.018 (for $z < 3$) by the time dependence used in the DESI modified-gravity analyses [28],

$$\mu(a) = 1 + \mu_0\,\frac{\Omega_{\mathrm{DE}}(a)}{\Omega_{\mathrm{DE},0}}, \qquad \mu_0 \simeq -0.20. \tag{14}$$

Gravity acting on matter clustering is weakened by about a fifth today, fading as dark energy becomes subdominant. The lensing response Σ is not fixed by the ladder (Open problem 4).

Early dark energy. Prediction 3 refers to the axion-like early dark energy of Poulin et al. [29], with potential $V(\phi) \propto [1-\cos(\phi/f_a)]^3$ (here ϕ is the EDE field, not the golden ratio φ) and peak fraction f_{EDE} at the critical redshift z_c.

4 Tests against current data

All code, data products and outputs are public (see Data and code availability).

4.1 Data and method

- **BAO.** 13 DESI DR2 measurements of D_V/r_d, D_M/r_d and D_H/r_d, with the full covariance [1].
- **CMB.** Planck 2018 distance priors (R, ℓ_A, ω_b) with their correlation matrix [30, 5], and, separately, DESI's official Markov chains computed with the full CMB likelihoods [3].
- **SNe.** Pantheon+ (1590 light curves with $z > 0.01$) [31], Union3 (22 binned magnitudes) [32], DESY5 (1829 SNe) [33] and the recalibrated DES-Dovekie (1820 SNe) [4], each with its full statistical and systematic covariance. The absolute magnitude is marginalized analytically:

$$\chi^2_{\mathrm{SN}} = \mathbf{r}^{\mathrm{T}} C^{-1} \mathbf{r} - \frac{(\mathbf{1}^{\mathrm{T}} C^{-1} \mathbf{r})^2}{\mathbf{1}^{\mathrm{T}} C^{-1} \mathbf{1}}. \tag{15}$$

- **Theory.** A flat universe with photons, $N_{\mathrm{eff}} = 3.046$, and one massive neutrino of 0.06 eV carrying $N_{\mathrm{eff}}/3$. The massive-neutrino density uses the fitting formula of Komatsu et al. [34] at all redshifts. The drag scale uses the DESI calibrated fit [2],

$$r_d = 147.05\,\mathrm{Mpc}\left(\frac{\omega_b}{0.02236}\right)^{-0.13}\left(\frac{\omega_m}{0.1432}\right)^{-0.23}\left(\frac{N_{\mathrm{eff}}}{3.04}\right)^{-0.1}. \tag{16}$$

Data	Model	$\chi^2_{\rm min}$	χ^2/dof	AIC	BIC	$n\sigma$ vs. free
DESI+CMB	ΛCDM	14.64	1.126	20.64	22.96	2.31
	φ-ladder	9.36	0.720	**15.36**	**17.67**	1.04
	w_0w_a	6.92	0.629	16.92	20.78	–
+Pantheon+	ΛCDM	1420.40	0.887	1426.40	1442.55	2.24
	φ-ladder	1414.03	0.883	**1420.03**	**1436.18**	0.51
	w_0w_a	1413.04	0.883	1423.04	1449.94	–
+Union3	ΛCDM	43.05	1.266	49.05	53.96	3.25
	φ-ladder	31.40	0.923	**37.40**	**42.31**	0.85
	w_0w_a	29.53	0.923	39.53	47.71	–
+DESY5	ΛCDM	1664.06	0.904	1670.06	1686.62	3.80
	φ-ladder	1646.88	0.895	**1652.88**	**1669.44**	0.28
	w_0w_a	1646.38	0.895	1656.38	1683.98	–
+DES-Dovekie	ΛCDM	1649.71	0.900	1655.71	1672.26	2.80
	φ-ladder	1639.71	0.895	**1645.71**	**1662.25**	0.30
	w_0w_a	1639.17	0.896	1649.17	1676.75	–

Table 1: Model comparison with compressed CMB information. The last column converts $\chi^2 - \chi^2_{w_0w_a}$ (2 degrees of freedom) into Gaussian σ. The φ-ladder has no free dark-energy parameters and the lowest AIC and BIC in every combination. Degrees of freedom subtract one per SN sample for the marginalized magnitude; $\chi^2/\text{dof} < 1$ for the large SN samples reflects their conservative error budgets. The $d = 3$ branch is 1.0–2.2σ from the free fit.

The distance priors of Ref. [30] were built with the decoupling redshift z_* of the Planck chains. We therefore use the Hu–Sugiyama fit [35] times a quadratic correction calibrated against CAMB [36], with residuals below 4×10^{-7}.

- **Validation against Planck and CAMB.** For Planck 2018 parameters our code gives $r_s(z_*) = 144.43\,\text{Mpc}$, $r_d = 147.09\,\text{Mpc}$ and $100\theta_* = 1.04110$, identical to the Planck values. For the φ-ladder best fit it agrees with CAMB to 0.001% in $H(z)$ and $D_M(z)$, 0.001% in R, 0.002% in ℓ_A, 0.03% in r_d (the accuracy of the DESI fit), and 0.23% in the shape of $f\sigma_8(z)$ up to $z = 2$.

- **Models.**
  - ΛCDM;
  - the hypothesis ("φ-ladder", Eqs. (5)–(6));
  - the alternative $d = 3$ branch, $1 + w_0 = x^3$;
  - w_0w_aCDM.

  All share the parameters $(\omega_{bc}, \omega_b, h)$; w_0w_aCDM adds two.

4.2 Late-time dark energy

Table 1 and Fig. 1 summarize the results. The φ-ladder lies 0.3–1.0σ from the free fit and is preferred over ΛCDM by $\Delta\chi^2 = 5.3$–17.2. In the same fits, ΛCDM is disfavored at 2.2–3.8σ, close to the published DESI values (2.8–4.2σ, full CMB); the difference comes from using compressed CMB information. With DES-Dovekie the free best fit is $(w_0, w_a) = (-0.822, -0.637)$, within 0.02 of the prediction. The DES-Dovekie analysis itself, with full CMB likelihoods, finds $w_0 = -0.803 \pm 0.054$ and $w_a = -0.72 \pm 0.21$ [4]; the prediction lies 0.1σ and 0.5σ from these values, one parameter at a time (no correlation coefficient is quoted, so we give no joint level). The best-fit φ-ladder backgrounds have $H_0 \simeq 67.1$–$67.2\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ and $\Omega_\text{m} \simeq 0.316$–$0.317$.

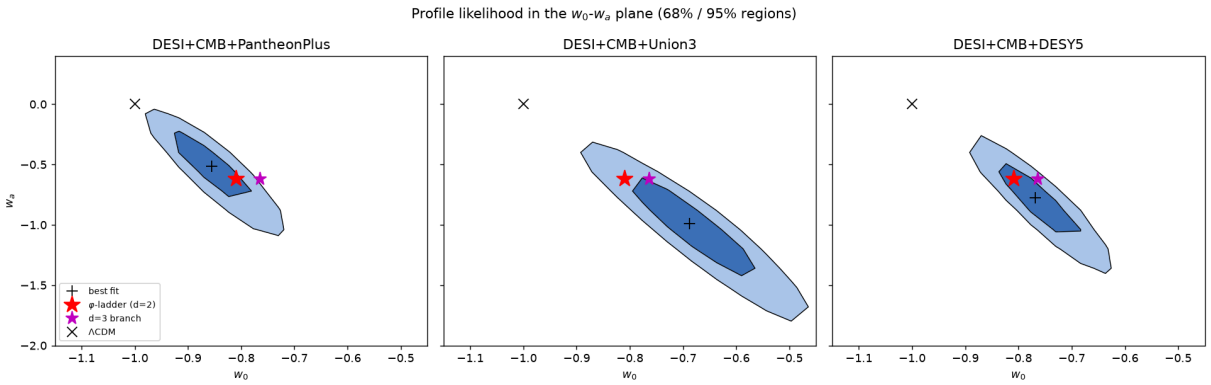


Figure 1: Profile likelihood in the w_0–w_a plane (68% and 95% regions) for DESI DR2 BAO + CMB distance priors + each SN sample. Red star: φ-ladder prediction. Magenta star: $d = 3$ branch. Cross: ΛCDM.

Official DESI DR2 chains	φ-ladder level	$n\sigma$	ΛCDM (DESI)
DESI+CMB (baseline)	85.3%	1.45	3.1σ
+Pantheon+	39.9%	0.52	2.8σ
+Union3	70.4%	1.04	3.8σ
+DESY5	40.4%	0.53	4.2σ
DESI+CMB, Planck PR3 plik	80.2%	1.29	
+DESY5, Planck PR3 plik	35.8%	0.47	
+DESY5, HiLLiPoP/LoLLiPoP	21.1%	0.27	
+DESY5 with $z > 0.1$	17.8%	0.23	
DESI BAO only (no CMB)	72.2%	1.08	
DESI BAO + DESY5 (no CMB)	10.7%	0.14	

Table 2: Position of $(-\varphi/2, -1/\varphi)$ in the official DESI DR2 w_0w_aCDM posteriors. The upper block is DESI's baseline; the lower block replaces the CMB likelihood, cuts low-redshift SNe, or drops the CMB entirely. The last column quotes DESI's own preference against ΛCDM [1]. The $d = 3$ branch lies at 1.7, 2.6, 1.4 and 1.6σ in the baseline chains. The non-crossing history of Appendix C lies outside the sampled region in every combination; a Gaussian estimate gives 4.0–5.3σ.

4.3 Official DESI posteriors with the full CMB likelihood

The DESI collaboration has released its DR2 Markov chains [3]. The paper's default CMB combination is:

- Planck PR3 low-ℓ TT and EE;
- Planck PR4 CamSpec high-ℓ TT, TE and EE;
- Planck PR4 and ACT DR6 lensing.

After discarding the first 30% of each chain as burn-in, we build a smoothed, weighted 2D histogram of (w_0, w_a). A point's credible level is the posterior mass in regions of higher density, i.e. the contour through it, converted to a two-sided Gaussian σ. Our posterior means reproduce the published DESI values to the quoted precision. For DESI+CMB+Pantheon+, for example, we find $w_0 = -0.838 \pm 0.055$ and $w_a = -0.617 \pm 0.208$, against the published -0.838 ± 0.055 and $-0.62^{+0.22}_{-0.19}$ (DESI quotes asymmetric 68% intervals for w_a) [1].

The prediction lies at 0.5σ (Pantheon+, DESY5) and 1.0σ (Union3) with supernovae, and at 1.45σ for DESI+CMB alone. It is equally consistent with alternative CMB likelihoods (0.27–

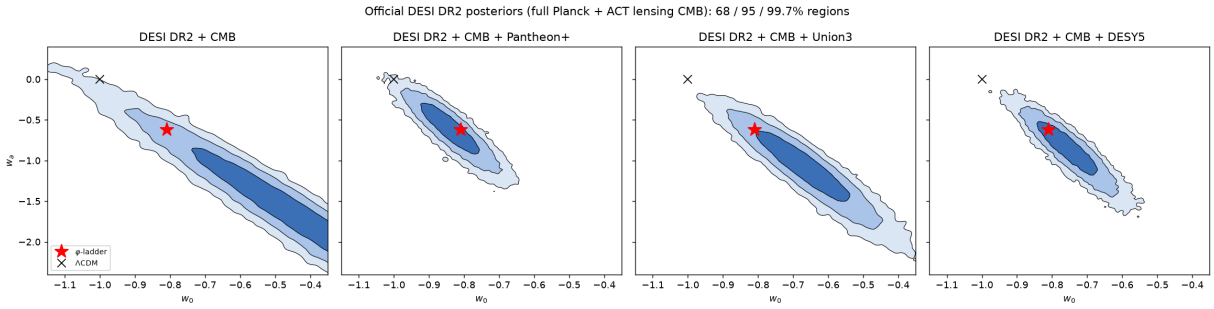


Figure 2: Official DESI DR2 w_0w_aCDM posteriors with the full Planck CMB likelihood and Planck + ACT lensing (68, 95 and 99.7% regions). Red star: φ-ladder prediction. Cross: ΛCDM.

Chains	z_p	w_p (DESI)	w_p (theory)	dev.	$w(0.3)$: dev.	$w(0.5)$: dev.
DESI+CMB	0.53	-1.024 ± 0.043	-1.023	0.02σ	-1.63σ	-0.31σ
+Pantheon+	0.31	-0.984 ± 0.025	-0.956	1.16σ	1.16σ	0.90σ
+Union3	0.39	-0.972 ± 0.032	-0.983	-0.33σ	-0.98σ	0.39σ
+DESY5	0.31	-0.954 ± 0.024	-0.954	0.00σ	-0.04σ	0.73σ

Table 3: Pivot and fixed-redshift tests in the official DESI chains. "dev." is (theory − DESI)/σ. The theory gives $w(0.3) = -0.9516$ and $w(0.5) = -1.0150$.

1.29σ), with a low-redshift SN cut (0.23σ), and without any CMB data (0.14–1.08σ), so the agreement is not driven by one dataset choice (Table 2, Fig. 2).

4.4 The best-measured redshift (pivot) and fixed-redshift tests

DESI constrains w most precisely at a pivot redshift z_p, where w is uncorrelated with w_a. Table 3 compares the prediction at each pivot and at the fixed redshifts of Eq. (11). At the pivot the prediction agrees to 0.00–1.16σ. The predicted crossing, $z = 0.447$, lies inside the range of pivot redshifts (0.31–0.53), so the theory, like the data, has w_p close to -1 at the best-measured redshift even though w_0 and w_a both differ from their ΛCDM values. This test uses the same chains as Table 2, so it is a sharper presentation rather than independent evidence.

4.5 Robustness of the dark-energy result

Two further tests, detailed in the appendices, check that the agreement is not an artifact of the CPL form or of the assumed early-universe physics:

- **Shape of $w(z)$ (Appendix A).** Against DESI's model-independent binned reconstruction, the predicted $w(z)$ deviates by 1.0–1.75σ, comparable to DESI's own best-fit CPL models (0.3–1.5σ).
- **Early-universe physics (Appendix B).** Without CMB data and with r_d free, the φ-ladder lies 0.1–1.1σ from the free fit; with CMB data and the predicted early dark energy included, it lies 0.2–1.3σ from it.

4.6 Letting the data measure x

We fit the one-parameter curve $(w_0, w_a) = (-1 + x^2/2, -x)$ with x free (Fig. 3). The data prefer the following values:

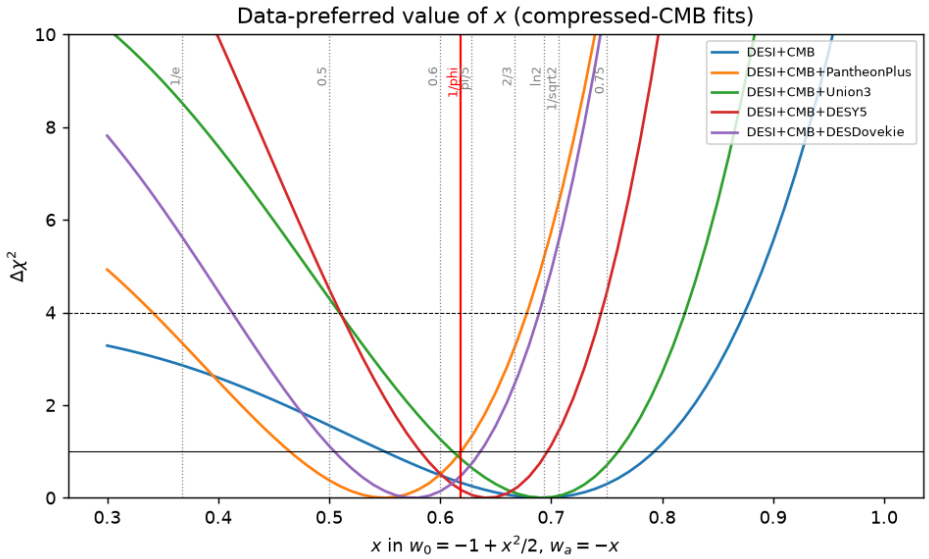


Figure 3: $\Delta\chi^2$ as a function of x on the ladder curve, for each data combination. Vertical lines mark simple constants; red marks $1/\varphi$. Horizontal lines: $\Delta\chi^2 = 1$ and 4.

Data	best x	68% range
DESI+CMB	0.692	[0.54, 0.80]
+Pantheon+	0.550	[0.46, 0.62]
+Union3	0.692	[0.61, 0.77]
+DESY5	0.643	[0.58, 0.70]
+DES-Dovekie	0.576	[0.50, 0.64]

The density of the official chains along the same curve gives consistent values (0.555–0.695). The five 68% ranges overlap only in $[0.61, 0.62]$, which contains $1/\varphi = 0.618$. Among nine simple constants ($1/e$, 0.5, 0.6, $1/\varphi$, $\pi/5$, 2/3, $\ln 2$, $1/\sqrt{2}$, 0.75), $1/\varphi$ is the only one inside every 68% range. It is, however, at the edge for Pantheon+ ($\Delta\chi^2 = 0.98$), and the nearby values 0.6 and $\pi/5$ miss only narrowly. The datasets share BAO and CMB information, so these ranges are not independent.

Conclusion: current data measure $x \approx 0.55$–0.69 and are fully consistent with $1/\varphi$, but they *cannot single out* φ from its neighbors. That is why the predictions are pre-registered.

4.7 Bayes factors

For a point nested in w_0w_aCDM, the Savage–Dickey ratio gives the Bayes factor directly as the posterior density at the point divided by the prior density. We use DESI's priors [1]: $w_0 \in U[-3, 1]$, $w_a \in U[-3, 2]$ and $w_0 + w_a < 0$, a prior area of 15.5. From the official chains, varying the histogram smoothing:

- **The φ point versus free (w_0, w_a):** B = 14–16 (DESI+CMB), 270–280 (+Pantheon+), 67–77 (+Union3) and 250–270 (+DESY5), so the fixed point is strongly favored over the two-parameter extension.
- **ΛCDM versus free (w_0, w_a):** B = 0.25–0.48, 2.2–2.5 and 0.01–0.07; for DESY5 the ΛCDM point lies outside the sampled region, so B is too small to estimate.

These numbers are dominated by the Occam penalty of the wide prior. The largest Bayes factor any single point can achieve, at the posterior peak, is 88, 464, 258 and 488 respectively. The φ point reaches 18–60% of these, and neighboring points on the curve (x = 0.60, $\pi/5$, 2/3)

score similarly. The Bayes factors therefore show that a simple fixed point near $1/\varphi$ is strongly preferred over free (w_0, w_a) and over ΛCDM in every combination ($B_\varphi/B_{\Lambda\mathrm{CDM}} \gtrsim 30$, 110 and 960 for DESI+CMB, +Pantheon+ and +Union3, and larger for +DESY5); they do not single out φ itself.

4.8 Look-elsewhere test

We fitted all 403 pairs (w_0, w_a) with $w_0 \in [-1.3, -0.3]$, $w_a \in [-2.5, 0.5]$ and $w_0 + w_a < 0$ drawn from the algebraic family $\{\pm c\, x^d + o : c \in \{1, \frac{1}{2}\},\ d = 1..4,\ o \in \{0, \pm 1\}\}$. This family contains every value the hypothesis uses, including both choices of α_d.

- The φ-ladder pair ranks 4th (Pantheon+), 15th (Union3), 4th (DESY5) and 2nd (DES-Dovekie) out of 403.
- For each sample separately, 10–19 pairs from the family lie within 1σ of the free best fit, so a match to one data set is not by itself significant.
- **Only one of the 403 pairs**, $(-\varphi/2, -1/\varphi)$, lies within 1σ for all three samples, whether the DES sample is DESY5 or DES-Dovekie.

The combinations share the BAO and CMB data and partially overlapping SNe, so they are not independent. The values were chosen after DESI DR2, so these results establish consistency, not confirmation.

4.9 Growth index

We use two independent sets of $f\sigma_8$ measurements.

SDSS-era data:

- 6dFGS at $z = 0.067$ [37];
- the SDSS MGS at $z = 0.15$ [38], using 0.53 ± 0.16 as tabulated in Ref. [39];
- BOSS DR12 at $z = 0.38, 0.51$ and eBOSS LRG at $z = 0.698$, with their joint covariance;
- eBOSS ELG at $z = 0.85$ and eBOSS QSO at $z = 1.48$ [39].

DESI DR1 data:

- six ShapeFit $f\sigma_{s8}$ measurements at $z = 0.30$–1.49 [40], with marginal errors from the published Gaussian covariances, treated as $f\sigma_8$ (valid when the fiducial sound horizon is close to the true one);
- the peculiar-velocity consensus $f\sigma_8(0.07) = 0.450 \pm 0.055$ [41].

The DESI DR2 Lyman-α full-shape analysis excluded its $f\sigma_8$ after mocks revealed a 10% bias [42], so it is not used. The two data sets overlap in volume and are analyzed separately. The growth factor is $\mathrm{d}\ln D/\mathrm{d}\ln a = \Omega_\mathrm{m}(a)^\gamma$, integrated from $z = 30$ in the φ-ladder background.

	Shape only (σ_8 free)		CMB-anchored	
Data	γ	$1/\varphi$ off by	γ	$1/\varphi$ off by
SDSS (6dF, MGS, BOSS, eBOSS)	$0.654^{+0.137}_{-0.130}$	0.3σ	$0.537^{+0.063}_{-0.059}$	1.3σ
DESI DR1 (FS + PV)	$0.568^{+0.120}_{-0.111}$	0.4σ	$0.536^{+0.072}_{-0.067}$	1.1σ

Table 4: Growth-index fits. In the CMB-anchored case the amplitude is fixed at $z = 30$ by Planck ($\sigma_8 = 0.8111 \pm 0.006$) and grown to low redshift by each model.

Five results stand out:

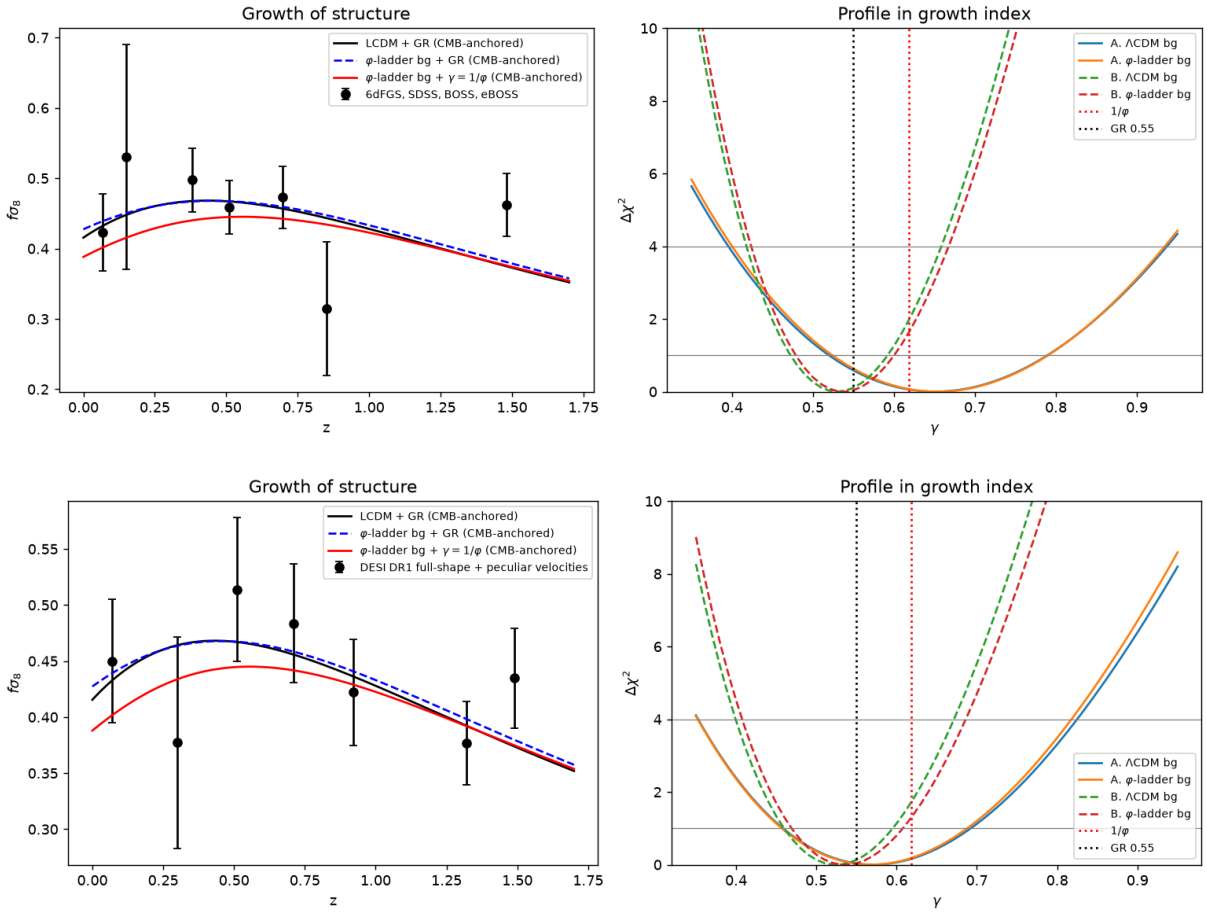


Figure 4: Top: SDSS-era $f\sigma_8$ data. Bottom: DESI DR1 full-shape and peculiar-velocity data. Left panels: data and CMB-anchored model curves. Right panels: $\Delta\chi^2$ profile in γ for shape-only (solid) and CMB-anchored (dashed) fits.

- The DESI shape-only result reproduces DESI's own measurement, $\gamma = 0.58 \pm 0.11$ [41], which validates the pipeline.
- Both data sets independently place $\gamma = 1/\varphi$ about 1σ from the CMB-anchored best fit, a mild preference for GR that is not significant.
- In terms of $\mu_0 \simeq -0.20$ (Sec. 3.3): the DESI full-shape analysis [28] finds, in a ΛCDM background, $\mu_0 = 0.11^{+0.44}_{-0.54}$ (DESI FS+BAO with BBN) and 0.05 ± 0.22 (with CMB, DES Y3 and DES Y5), at 0.6σ and 1.1σ from the prediction. In a w_0w_aCDM background, closer to the φ-ladder, the same data give $\mu_0 = -0.24^{+0.32}_{-0.28}$ and $\Sigma_0 = 0.006 \pm 0.043$. That agrees with the prediction within 0.2σ, but it is also consistent with GR, so it does not discriminate.
- The implied clustering amplitude is $S_8 = 0.812$. Here a lower $\sigma_8 = 0.790$, from the weaker growth, is partly offset by a higher $\Omega_m = 0.317$ in the φ-ladder background. The value agrees with KiDS-Legacy (0.815, 0.2σ) [7] but lies 2.1σ above DES Y3 (0.776 ± 0.017) [6].
- The result of Nguyen et al. [43], which uses additional probes, is $\gamma = 0.633^{+0.025}_{-0.024}$, 0.6σ from $1/\varphi$.

Appendix C shows that elastic momentum exchange between dark matter and dark energy cannot produce this suppression in a background that crosses $w = -1$.

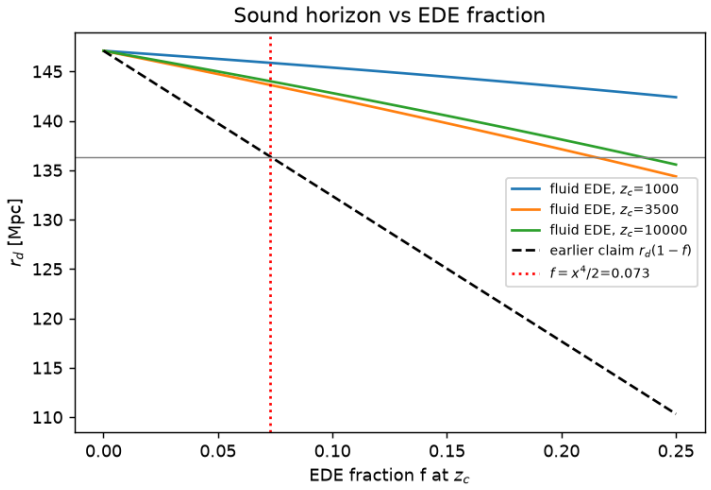


Figure 5: Drag-epoch sound horizon versus peak early-dark-energy fraction for three critical redshifts z_c, compared with the incorrect scaling $r_d(1-f)$ (dashed).

4.10 Early dark energy and H_0: a corrected claim

An earlier version of this proposal derived $r_d = 147.09\,(1 - f_{\rm EDE}) = 136.36\,\mathrm{Mpc}$ and hence $H_0 = 67.40/(1 - f_{\rm EDE}) = 72.70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. This assumes that the sound horizon shrinks by exactly the energy fraction, which is incorrect. In the standard fluid approximation to the axion-like early dark energy of Ref. [29] ($n = 3$, so $w_n = (n-1)/(n+1) = 1/2$):

- $f_{\rm EDE} = 0.073$ reduces r_d by only 0.8–2.4% for $z_c = 10^3$–10^4 (Fig. 5);
- reaching 136.36 Mpc would require $f_{\rm EDE} \approx 0.21$;
- the implied H_0 ranges from 69.0 (holding $H_0 r_d$ fixed) to $76.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (holding θ_* and ω_c fixed), and only a full Boltzmann-code fit can settle it.

We therefore *do not* predict H_0. Current 95% upper limits are:

- $f_{\rm EDE} < 0.070$ from Planck CMB and lensing, and < 0.091 adding DESI BAO [46];
- < 0.065 (Planck NPIPE) and < 0.092 (Planck + ACT DR6), both with CMB lensing and Pantheon+ and without DESI BAO or the SH0ES prior; adding DESI DR2 BAO to the second gives < 0.109 [47];
- < 0.12 for a combination including ACT DR6 [48].

Prediction 3 is therefore already under pressure. Including this early dark energy does not change the agreement of Prediction 1 with the data (Appendix B).

5 Pre-registered predictions and falsification criteria

The following were time-stamped on 19 October 2026 (Zenodo DOI: 10.5281/zenodo.23481070). The SHA-256 hash of the prediction record is

70b9cfb23fc1e0d119ebd81aed270933960fa52256f89f505f799bb3a81767f2

What counts as a test. DESI's DR2 results use observations from 14 May 2021 to 9 April 2024 [1], and the predictions were chosen after them. The first DESI test is therefore any cosmology analysis that includes observations made after 9 April 2024. No release date for such results has been announced. Because future DESI analyses also contain the DR2 data, part of their agreement is built in; fully independent tests come from other probes (Euclid growth measurements, new supernova samples and CMB experiments).

Prediction 1. $(w_0, w_a) = (-\varphi/2, -1/\varphi) = (-0.80902, -0.61803)$, equivalently $w_a = 1 + 2w_0$ with $w_0 = -\varphi/2$. Hence $w = -1$ at $z = 1/\sqrt{5}$, $w(0.3) = -0.9516$ and $w(0.5) = -1.0150$.

Falsified if any of the following holds in the first DESI analysis including post-DR2 data:

- the point lies outside the 3σ region ($\Delta\chi^2 > 11.83$) of the w_0–w_a constraint from BAO + CMB alone, or from BAO + CMB + SNe for at least two of the principal SN compilations;
- $w(0.3)$ or $w(0.5)$ differs from the prediction by more than 3σ in at least two of these combinations.

Supported if the point stays within 2σ in all combinations while ΛCDM is excluded at more than 3σ.

Prediction 2. $\gamma = 1/\varphi = 0.618$, constant for $0 \leq z \leq 1$.

Falsified by a measurement with $\sigma_\gamma \leq 0.03$ lying more than 3σ from 0.618.

Prediction 3. $f_{\rm EDE} = \frac{1}{2}x^4 = (5 - 3\varphi)/2 = 0.07295$ (axion-like EDE, $n = 3$, z_c free).

Falsified if excluded at 99.7%.

Derived, not independently tested: $S_8 \approx 0.81$ (0.812 in the best-fit background of Sec. 4.9; approximate, because it depends on the background fit). H_0 is not predicted.

6 Open problems

Open problem 1 (The bridge)**.** Sec. 2 proves that $1/\varphi$ characterizes maximal non-resonance. Why the cosmological dark sector should be governed by this property is postulated (Hypothesis 1), not derived.

A conjectural route. In the minisuperspace (Hamiltonian) description, a closed Friedmann universe with one scalar field has two degrees of freedom (a, ϕ): a four-dimensional phase space on which the Friedmann equation defines a three-dimensional constraint surface, which is exactly the setting of the KAM results of Sec. 2. This system is known to be chaotic [49, 50, 51]. For a spatially flat universe with a single field, the dynamics reduces to the two-dimensional autonomous system $(\phi, \dot{\phi})$, which cannot be chaotic. However, a single canonical scalar field cannot cross $w = -1$ [25]; the simplest models that do, quintom models, use two fields [22], and two-field cosmological dynamics can be chaotic [52]. We therefore conjecture that the dynamics of a (multi-field) dark sector is organized by its most robust non-resonant barriers, the golden ones. Showing this, and showing that it produces the ladder of Hypothesis 2, is open. In particular, why the early-dark-energy fraction should be a $d = 4$ term is not explained.

Open problem 2 (The ladder)**.** The weights α_d and the assignment of d to each observable (Hypotheses 2 and 3) are postulates. The rival $d = 3$ branch ($w_0 = -0.764$) fits worse but is not excluded by any principle.

Open problem 3 (The value of w_a)**.** $w_a = -1/\varphi$, or equivalently the crossing position $1 - a_{\rm cross} = x/2$, is postulated, not derived. A dynamical model that yields both w_0 and w_a from one equation of motion is needed.

Open problem 4 (Mechanism)**.** The effective embedding of Sec. 3.3, PPF dark energy plus $\mu_0 \simeq -0.20$, is consistent but phenomenological. It also leaves the lensing response Σ open; current data require $\Sigma_0 \approx 0$ to within about 0.05 [28]. Pure momentum exchange cannot produce the growth suppression across the crossing (Appendix C). A candidate is energy exchange,

$$\dot{\rho}_c + 3H\rho_c = -Q, \qquad \dot{\rho}_{\rm DE} + 3H(1+w)\rho_{\rm DE} = +Q. \tag{17}$$

An observer who assumes $\rho_c \propto a^{-3}$ infers $\rho_{\rm DE,eff} = \rho_{\rm DE} + \Delta$, with $\Delta = \rho_c - \rho_{c,0}a^{-3}$, and

$$1 + w_{\rm eff} = \frac{(1+w)\,\rho_{\rm DE} + \Delta}{\rho_{\rm DE} + \Delta}. \tag{18}$$

An apparent phantom past ($w_{\rm eff} < -1$) therefore needs $\Delta < 0$: dark matter must have been less dense in the past than a^{-3} implies, so energy must flow *from dark energy to dark matter* ($Q < 0$) [53].

We tested the simplest forms, $Q = -\xi H \rho_{\rm DE}$ and $Q = -\xi H \rho_c$, with constant $w = -\varphi/2$, at the level of the background:

- The crossing can be placed at $z = 1/\sqrt{5}$, with $\xi = 0.34$ and 0.58 respectively, and $w_{\rm eff}$ then follows the prediction to within 0.04 up to $z = 0.5$.
- Beyond that, $w_{\rm eff}$ falls far too fast (-1.73 and -2.00 at $z = 1$, against -1.12 predicted), and the inferred dark-energy density turns negative by $z = 1.6$ and 1.4.

These simple forms are therefore excluded as the mechanism. A viable model would need a time-dependent coupling or equation of state, and its growth rate (μ_0) depends on how momentum is transferred, which we have not modeled.

Open problem 5 (Beyond cosmology)**.** The intuition that recursive, non-resonant partitioning of phase space also organizes protein folding or neural action selection is not expressed here as a quantitative prediction, and is not claimed. Any such claim would need to compete with established explanations:

- **Protein folding.** Energy-landscape ("folding funnel") theory resolves Levinthal's paradox with biased, multi-pathway search [54].
- **Readiness potential.** The pre-movement potential has been modeled as the accumulation of ongoing neural noise rather than a pre-made decision [55].

A testable version would have to predict, for example, folding-time scaling or readiness-potential timing.

7 Conclusion

We have reduced an intuition, that trajectories in higher-dimensional phase space must avoid one another and that this organizes nature around the golden ratio, to two separate parts:

- **Mathematics:** proven results (non-merging of distinct states; $1/\varphi$ as the global attractor of recursive self-similar partitioning and the most non-resonant number), plus well-supported numerical evidence that golden tori are the most robust KAM barriers.
- **A falsifiable hypothesis:** the dark-sector parameters that differ from ΛCDM take ladder values $\alpha_d x^d$.

The hypothesis fixes $w_0 = -\varphi/2$ and $w_a = -1/\varphi$ without free parameters. With compressed CMB information this pair lies 0.3–1.0σ from the best free fit, including the recalibrated DES-Dovekie sample, and has the lowest AIC and BIC in every combination. In DESI's official posteriors it lies at 0.5–1.0σ when supernovae are included, and it is robust to the choice of CMB likelihood. It is the only member of its 403-member algebraic family within 1σ of all three supernova samples. The agreement does not depend on the CMB or on the predicted early dark energy, and the full predicted $w(z)$ matches DESI's binned reconstruction about as well as DESI's own best-fit model.

When x is fitted freely, the data give $x \approx 0.55$–0.69, consistent with $1/\varphi$ but unable to single it out. The fixed-point Bayes factors favor a simple point near $1/\varphi$ over free (w_0, w_a) and over ΛCDM, but not φ specifically.

Each prediction has an established effective form: PPF dark energy for the crossing, $\mu_0 \simeq -0.20$ for growth, and axion-like early dark energy. The status of the other claims:

- The growth index $1/\varphi$ is mildly disfavored, at about 1σ, by two independent $f\sigma_8$ data sets.
- The early-dark-energy fraction is in tension with some CMB limits.
- An earlier H_0 claim is withdrawn.
- Neither momentum exchange nor the simplest energy-exchange couplings can reproduce the predicted growth suppression and $w(z)$ together.

The predictions are time-stamped. DESI data beyond DR2, Euclid and next-generation CMB experiments will decide them.

Data and code availability

All analysis code and outputs are archived on Zenodo (DOI 10.5281/zenodo.23481070) and on GitHub (`https://github.com/letsbuild9/golden-ratio-dark-energy`):

- likelihoods, fits and validation: `borse_fit.py`, `borse_calibrate_zstar.py`, `borse_checks.py` and `borse_camb_check.py`;
- official DESI chains and robustness tests: `borse_desi_chains.py`, `borse_v2_tests.py`, `borse_v2b_tests.py` and `borse_binned_calibration.py`;
- growth and mechanism tests: `borse_growth.py`, `borse_mu.py`, `borse_model.py` and `borse_energy_exchange.py`;
- verification: `borse_symbolic.py` (exact symbolic proofs), `verify_paper.py` (an automated check that every number quoted here matches the outputs) and a ledger recording the source of every external value;
- `lock_predictions.py`, which writes the time-stamped prediction record and its hash.

Public data were obtained from:

- the DESI DR2 BAO cosmology chain release [3];
- the DESI DR1 full-shape [40] and peculiar-velocity [41] analyses;
- SDSS DR16 [39];
- the Pantheon+, Union3, DES-SN5YR and DES-Dovekie releases, via the Cobaya `bao_data` and `sn_data` repositories.

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A Model-independent shape: DESI's binned $w(z)$

CPL is only a local parametrization. DESI also fits w as piecewise constant in $z = 0$–0.7, 0.7–1.4, 1.4–2.1 and 2.1–10, with $w = -1$ beyond [3]. DESI's implementation smooths the steps between bins (fixed smoothing parameter $s = 0.02$ in the released configuration); we treat the bins as

sharp. We compare the official binned posteriors with the predicted $w(z)$ averaged over each bin, uniformly in z (Table 5). Uniform averaging is only an approximation to what a piecewise-constant fit returns, so we calibrate it by applying the identical averaging to each data set's own w_0w_aCDM posterior-mean model.

Official binned chains	w (0–0.7)	w (0.7–1.4)	w (1.4–2.1)	φ-ladder	DESI's CPL
DESI+CMB	-0.840 ± 0.073	-1.485 ± 0.204	-1.209 ± 0.772	1.18σ	0.29σ
+Pantheon+	-0.919 ± 0.030	-1.384 ± 0.172	-1.255 ± 0.751	1.01σ	1.53σ
+Union3	-0.862 ± 0.042	-1.454 ± 0.181	-1.209 ± 0.753	1.75σ	0.91σ
+DESY5	-0.894 ± 0.029	-1.362 ± 0.168	-1.366 ± 0.746	1.63σ	1.52σ

Table 5: DESI DR2 binned $w(z)$ versus bin-averaged predictions; the predicted averages are -0.959, -1.123 and -1.201. The last two columns give the joint deviation over the first three bins (full covariance, 3 degrees of freedom) for the φ-ladder and for each data set's own CPL posterior-mean model.

The predicted $w(z)$ is consistent with the binned reconstruction at 1.0–1.75σ, comparable to DESI's own best-fit CPL models (0.3–1.5σ). Both CPL curves are less negative than the data between $z = 0.7$ and 1.4 and, except for DESI's own fit to DESI+CMB, more negative below $z = 0.7$. This appears to be a limitation of the CPL shape rather than of the golden-ratio values.

B Joint check: dark energy with the predicted early dark energy

The dark-energy fits of Sec. 4 assume standard early physics, but the hypothesis also predicts $f_{\rm EDE} = 0.073$, which shortens the sound horizon. We test whether Prediction 1 survives this in two ways.

(a) Independent of early physics. DESI BAO alone, or with each SN sample, with r_d left free (only $h\,r_d$ enters, so any early-universe change is absorbed) and no CMB. The φ-ladder lies 1.07 (BAO only), 0.76 (+Pantheon+), 0.63 (+Union3), 0.10 (+DESY5) and 0.25σ (+DES-Dovekie) from the free fit, while ΛCDM is disfavored at 1.7–3.3σ. This agrees with the official no-CMB chains of Table 2.

(b) With CMB and early dark energy (indicative). We add the axion-like fluid with $f_{\rm EDE} = 0.073$ to $H(z)$ for $z_c = 10^3$, 3.5×10^3 and 10^4, and rescale r_d and $r_s(z_*)$ by the ratio of the sound-horizon integrals with and without it (r_d shrinks by 0.8–2.3%). Across all z_c and data combinations the φ-ladder stays 0.2–1.3σ from the free fit, ΛCDM is disfavored at 2.0–4.0σ, and H_0 rises to 68.1–70.4 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. The compressed CMB priors assume standard early physics, so this case is indicative only; a full Boltzmann-code analysis with the CMB likelihood is needed for a definitive statement.

Conclusion. At current precision, Prediction 1 does not depend on whether the predicted early dark energy is present.

C Momentum exchange as the growth mechanism: a negative result

A natural mechanism for suppressing growth without altering the background is elastic dark-matter–dark-energy scattering [44], related to the pure-momentum-transfer coupling of Pourtsidou, Skordis & Copeland [45]. In the sub-horizon, quasi-static limit, with $N = \ln a$,

$$\delta_c'' + \left(2 + \frac{\mathrm{d}\ln H}{\mathrm{d}N} + \xi\,(1+w)\,\Omega_{\rm DE}\frac{H}{H_0}\right)\delta_c' = \tfrac{3}{2}\left(\Omega_c\delta_c + \Omega_b\delta_b\right), \tag{19}$$

$$\delta_b'' + \left(2 + \frac{\mathrm{d}\ln H}{\mathrm{d}N}\right)\delta_b' = \tfrac{3}{2}\left(\Omega_c\delta_c + \Omega_b\delta_b\right), \tag{20}$$

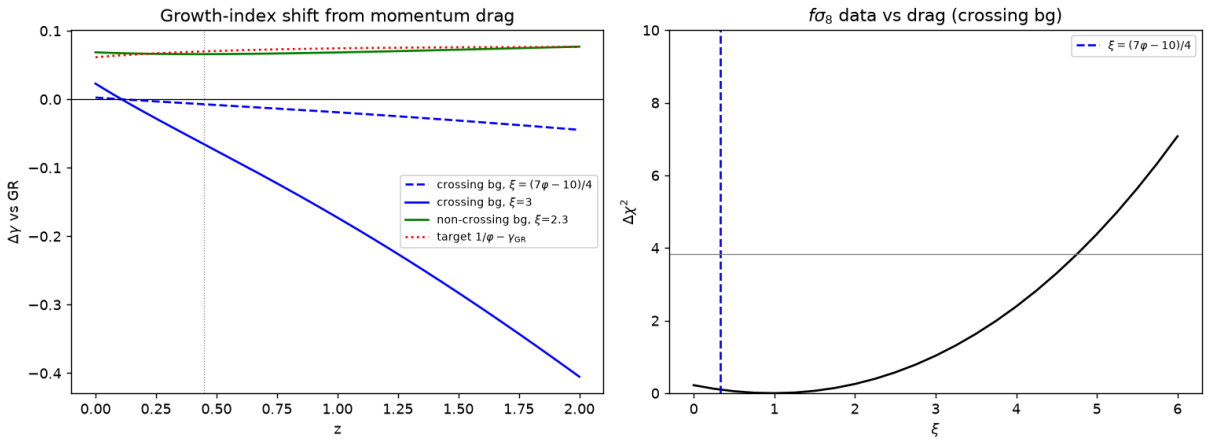


Figure 6: Left: growth-index shift produced by dark-matter–dark-energy momentum exchange. Blue: phantom-crossing φ-ladder background. Green: non-crossing background. Dotted red: target, $1/\varphi - \gamma_{\rm GR}(z)$. Right: $f\sigma_8$ constraint on ξ in the crossing background.

with $\xi = (\sigma_D/m_c)\, c\, \rho_{\rm crit,0}/H_0$. Because the drag is proportional to $1+w$, it *reverses sign* above $z_{\rm cross}$. Integrating in the φ-ladder background (Fig. 6), with a target suppression equal to that implied by $\gamma = 1/\varphi$ (Prediction 2), i.e. $\gamma - \gamma_{\rm GR} \approx 0.06$–$0.08$ for $0 \le z \le 1.5$:

- **No coupling $\xi \ge 0$ produces the required suppression.** The target is a 2.6% lower growth factor at $z = 0$. Instead, growth is *enhanced*, by +0.45% for $\xi = (7\varphi - 10)/4 \approx 0.33$ and by +4.2% for $\xi = 3$.
- **A non-crossing history fixes the sign but is excluded.** A CPL history with the same w_0 never crosses -1 only if $w_a \ge -(1+w_0) = -0.191$. At this boundary a coupling $\xi \approx 2.3$ ($\sigma_D/m_c \approx 36\,\mathrm{b/GeV}$) gives an almost constant $\Delta\gamma = 0.066$–0.072 for $0 \le z \le 1.5$, close to the 0.059–0.069 required in that background. However, this history is excluded at 4.2–4.9σ by the data of Table 1, and at 4.0–5.3σ (Gaussian estimate) in the official DESI posteriors.

Momentum exchange can therefore produce the growth suppression only in a background that the data exclude.

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